

A Computable Calculus of Conjecture, Criticism, and Background

Abstract. We present a calculus in which every epistemic judgment is computable: statuses are labels of a Dung grounded extension extended by a support pass, all evaluation is resource-bounded and deterministic, and the entire state is a replayable fold over an append-only event log. The calculus is critical-rationalist throughout: demarcation is explanatory, not merely predictive; there is no induction, no confirmation relation, and no probability over the truth of any claim; there is no authority and no final state. Its central contribution is a **two-axis epistemic state**. Alongside *status* (truth-standing under criticism) it computes *standing* (role in the economy of generation: whether an artifact is framed, or does the framing). The second axis realizes the doctrine of background knowledge — methodologically privileged for the occasion, epistemically privileged never — and makes representable the ordinary condition of mature science: a framework that is refuted and still framing. The same axis is what the logic of experimental tests requires, since refutation is unilateral but rational theory-choice is intrinsically comparative. Promotion into background is earned by measured reach and adjudicated like any conjecture; background is refuted by expectation-violation through the ordinary warrant machinery; its fall is a distinct, adjudicated event whose consequence is a mechanical criticism-cascade through every problem posed in its terms. Knowledge itself receives a computable characterization — unrefuted, hard to vary, reaching — as a derived view that can steer attention and never adjudicate. Every mechanism reduces to untyped artifacts, attack and dependence edges, problem criteria, and deterministic render policy; the calculus formalizes in full the

selective half of an evolutionary process whose generative half it deliberately leaves open, while affirming the computability of the whole.

1. Epistemological invariants

The calculus is the enforcement of eleven commitments.

Everything after this section is machinery discharging them; each is cited where discharged.

P1 (Fallibilism). No element — content, tests, evidence, rules, measures, the calculus's own procedures — is certain, justified, or final. Consequence: every status admits an exit; nothing is ever verified; the machine never certifies a terminal state.

P2 (No induction). Theories are not derived from observations. Generation is conjecture, unconstrained by the calculus; observation enters solely as criticism. Consequence: the calculus contains **no confirmation relation**. There is no edge by which evidence supports a claim; evidence can only ground attacks. Acceptance into the record is survival, not support.

P3 (Criticism is the sole selector). Epistemic state changes only through registered criticism and its defeat. A judgment that is not an attack, or the defeat of an attack, moves nothing.

P4 (Explanatory demarcation). Knowledge-apt content is an *explanation that forbids something*: an account of a mechanism whose assertions rule out states of affairs. Both halves are necessary. Content that forbids nothing has no attack surface and is outside the game; content that forbids without explaining — prophecy, oracle, bare correlation — dies for its explanatory emptiness *before and without testing*, since most bad ideas are rightly rejected unexperimented. Empirical falsifiability is the special case in which a forbidden thing is an observation.

P5 (Theory-ladenness of observation). Observations are themselves conjectures. Acceptance of a basic statement is a revisable decision; an observation refutes only while it itself survives criticism. Science rests on piles driven into a swamp, and the piles are inspectable.

P6 (No authority). No source enjoys status privilege — not the user, not the rules, not the calculus. Rule-objects are artifacts inside the game and attackable there. Authority may exist only as *attention* — what is rendered, in what order — never as status.

P7 (Problems first). Inquiry runs $P_1 \rightarrow$ tentative theory $\rightarrow$ error elimination $\rightarrow P_2$. There is no conjecture without a problem, and the growth of knowledge is the succession of problems, not the accumulation of answers.

P8 (Error preservation). Refuted knowledge is retained as *refuted*. Error correction requires the error's record; deletion is forgery of the growth sequence.

P9 (Hard to vary). An explanation whose components can be freely substituted while its performance is preserved is bad even if unrefuted. Variability is itself criticizable.

P10 (Unbounded conjecture). The space of proposable conjectures is open: every unrefuted artifact has proposable successors with strictly extended attack surfaces. The game cannot halt by exhaustion — though a *conditioned* generator's effective reach can collapse while in-principle reach stays unbounded, and the calculus must police the gap.

P11 (Background). Every test and every problem is posed against background knowledge held unproblematic *for the occasion*. This privilege is methodological and revocable: background directs the searchlight — it shapes what is conjectured and which problems are posable — while remaining, in principle, the most exposed knowledge in the system, since its commitments stand open

indefinitely. Two consequences frame the second half of this paper: background must be renderable as the standing frame of generation without ever being insulated from criticism; and when background falls, the problems posed in its terms — not merely the answers — come up for review.

P11 is where this calculus parts company with its neighbours. Lakatos protected the hard core and directed criticism at a belt; here the core pays for its position with *more* exposure, not less. Kuhn observed that paradigms persist through anomaly and concluded the persistence was extra-rational; here persistence-through-refutation is a computed, criticizable state with its wounds rendered in public.

2. Computability discipline

C1 (Determinism and replay). The append-only event log is the sole ground truth. Every verdict, label, measure, and rendered prompt is a pure function of the log; replaying the log reproduces the state byte-for-byte. Wall-clock time never enters a verdict.

C2 (Budgeted decidability). Every evaluation runs under a declared, finite, deterministic budget (step counts, item counts). Verdicts are total over {pass, fail, overrun}; `overrun` means "unobtainable within the declared budget," never "the machine was slow." Budgeting is what keeps Rice's theorem at bay: the calculus never asks an undecidable question.

C3 (Untypedness). Artifacts carry no kind field. All dispatch is on interface structure — what an artifact commits to, attacks, and depends on. A type would be an unattackable classification: status privilege by construction, violating P6, and a stored judgment, violating P1. Roles that look like types (standard,

evidence, precedent, frame) are patterns of structure and render policy, recoverable by query.

C4 (Computed, never stored). Status and standing are materialized views over the log, recomputable at any historical sequence number. Nothing is deleted (P8); a historical view is physically read-only.

C5 (Measures never adjudicate). Numeric signals — hardness-to-vary, reach, isolation, diversity — influence the graph through exactly three channels: (a) spawning problems, (b) being packaged as budgeted commitments whose *fail* verdicts generate warranted attacks, (c) steering attention (scheduling, rendering, budgets). They never enter label computation. A metric is not a criticism (P3).

C6 (No credence). No probability ever attaches to the truth of an artifact. Probabilistic support is not inductive support: every evidential rise in a hypothesis's probability decomposes into a deductively entailed component and a countersupported remainder (the Popper–Miller decomposition), so a probability calculus can carry no epistemic weight that the attack graph does not already carry. Corroboration is a record of criticism survived, never a magnitude of confidence; statistics of the generator's output distribution are properties of the generator, never of any conjecture's truth; and no prediction whose outcome depends on the future growth of knowledge is assigned a probability, since the growth of knowledge is precisely what no distribution ranges over. Where probability appears *inside* content — a stochastic model and its forecasts — it is content, adjudicated like any other.

3. Ontology

Definition 3.1 (Content). Content is opaque bytes plus a codec (utf8, json, csv, numeric, code). Meaning is imposed by conjecture and checked by program.

Definition 3.2 (Artifact). An artifact is

```
a = { id, content_ref, codec, interface, provenance }
interface = { commitments : [κ-id], refs : [{target,
role}] },
role ∈ { dependence, mention, evidence }
id = hash(canonical(content_ref, codec, interface))
```

Warrant carriage is an explicit append-only relation $\text{carry} \subseteq A \times W$; a later event may add a pair without changing the artifact's identity. `dependence` refs contribute support edges to `dep` (kept acyclic); `mention` refs are non-load-bearing; `evidence` refs are permitted on warrant validity nodes and declare load-bearing evidence (Closure 3, below). Provenance records the generating role and event, and is **epistemically inert**: it may steer attention, never labels.

Definition 3.3 (Commitment). A commitment $\kappa = \langle \text{eval}, \text{budget}, \text{observation_valued} \rangle$ where `eval` is a program, a decidable predicate, or a rubric reference (§11). Its verdict on content c is

$$V(\kappa, c) = U^{\wedge\{\leq\beta\}}(\tau_{\kappa}, c) \in \{ \text{pass}, \text{fail}, \text{overrun} \}$$

— extensional, budgeted, decidable (C2). An observation-valued commitment with no covering evidence artifact spawns a research problem; evidence sealed under holdout (§11) does not count as covering before its reveal.

Remark (commitments are counterfactuals). A commitment asserts that a transformation of the world — its forbidden case obtaining while its carrier stands unrefuted — is impossible. The interface is therefore the artifact's content in counterfactual form:

a specification of which tasks are possible and which impossible, and the calculus adjudicates nothing else. What an artifact *is*, epistemically, is what it rules out.

Definition 3.4 (Warrant). A warrant is a contentful attack: demonstrative (a commitment, its fail verdict, and the trace) or argumentative (a case). Every warrant carries a **validity node** v — an artifact asserting the test was sound and relevant. A bare verdict is never an edge.

Closure rules (enforced during `att` construction):

1. *Validity closure.* Any attacker of v attacks the warrant, hence the carrier's attack edge.
2. *Case-law closure.* The v of any rubric-derived warrant refs the standard it applied; every registered attacker of that standard attacks v . Refute a standard $\Rightarrow$ every verdict under it falls $\Rightarrow$ every target reinstates, computed in one pass.
3. *Evidence closure.* A v grounded in recorded evidence refs it; attackers of the evidence **or of anything in its transitive dependence lineage** attack v . Invalidating a source reinstates the target with no rule outside `att / dep`. This is P5 made mechanical: the faulty instrument is an attack on the observation, and its success is an ordinary reinstatement.

Definition 3.5 (Problem). $\pi = \langle \text{description, criteria, provenance} \rangle$. Criteria are commitment schemas instantiated per candidate; the **root battery** is pinned into every problem: internal-consistency checks and the two halves of explanatory demarcation (P4) — the attack surface is nonempty, and, for empirical scopes, a mechanism is present and *load-bearing*, in that role-level substitution or deletion of it flips verdicts. A mechanism that can be swapped freely is decoration, and its carrier is refuted by program without an experiment being run. Provenance records the spawn trigger and, when the problem is posed under active background, the field

```
provenance.frame : [ frame-assertion ids ]  
(§9.8)
```

written deterministically at registration, editable by the registrant at pose, immutable thereafter (C4). Like all provenance it is epistemically inert.

Definition 3.6 (State). $S = (A, \Pi, \text{carry}, \text{att}, \text{dep}, \text{addr}, \text{status}, \text{standing}, \text{measures})$ — a materialized view, recomputable at any log position.

4. Well-formedness

A state is well-formed iff every carried warrant names registered objects; every attack edge derives from a carried warrant; every criterion is a commitment schema; the three closures hold; `dep` is acyclic; and every rubric-derived warrant's trace contains a conforming trial transcript (§11). All transition rules preserve well-formedness.

5. Dynamics

Transition rules.

Rule	Enabling condition	Effect
Conj	$\Pi \neq \emptyset$; a problem π selected	register $a = \gamma(\pi, S)$ with interface attached; $\text{addr} += (a, \pi)$. No problem, no conjecture (P7).
Crit	target a ; a valid warrant w	register the critic if new; $\text{carry} += (k, w)$; derive $\text{att} += (k, a)$

Rule	Enabling condition	Effect
Adj	after any registration	recompute the two-pass labels (§6)
Spawn	any trigger below	register a new problem with provenance
Refl	always	the calculus's own rules, standards, render policies, and guard procedures are registered artifacts – attackable (P6)

Spawn triggers (exhaustive): failed verdict $\Rightarrow$ successor problem; ≥ 2 surviving rivals for one $\pi \Rightarrow$ discrimination problem; unrefuted artifact with low hardness-to-vary $\Rightarrow$ remove-arbitrariness problem; reach event $\Rightarrow$ explanation-debt problem; reach events for one artifact spanning $\geq K_{\text{frame}}$ distinct lineages over a coherent scope $\Rightarrow$ **promotion problem** (§9.4); a frame assertion leaving unrefuted standing $\Rightarrow$ **premise-orphan problems**, lazily materialized (§9.8); uncovered observation-valued commitment $\Rightarrow$ research problem; critic-gaming signal $\Rightarrow$ audit-the-critic problem; isolation above floor $\Rightarrow$ connection problem; unrefuted artifacts on overlapping problems with no declared relation $\Rightarrow$ integration problem.

Registration guards are warrant-validity conditions, not censorship: they suppress noise, never criticism (P3). *Anti-relapse*: a candidate hash-identical to a refuted artifact, or whose verdict-vector over the active battery matches a refuted prior's, is blocked **unless** it carries a warrant against that prior's refuter; near-duplicates of unrefuted artifacts are never blocked (blocking them would be a diversity gate adjudicating, violating C5). *Trial guard*: a rubric verdict registers as a warrant only downstream of an adversarial transcript (critic's case, defender's answer, ruling citing

a `decisive_point` that resolves to a real element of the exchange), order-swap consistency for comparative modes, and paraphrase spot-checks. Blocked rulings are logged; a streak of blocks is itself a spawn signal.

6. Adjudication

Labels are computed in two passes; inputs are `att` and `dep` **only** (C5, C6).

```
# Pass 1 – attack (Dung grounded extension: unique,
skeptical, polynomial)
F(X) = { a ∈ A : ∀(b,a) ∈ att, ∃c ∈ X with (c,b) ∈
att }
G    = least fixed point of F from ∅
labelo(a) = unrefuted  if a ∈ G
           refuted     if ∃b ∈ G with (b,a) ∈ att
           suspended   otherwise

# Pass 2 – support (over the dep DAG in topological
order)
supported(a) = ∀(a,b) ∈ dep : final(b) = unrefuted
final(a) = unrefuted          if
labelo(a)=unrefuted ∧ supported(a)
           suspended_unsupported  if
labelo(a)=unrefuted ∧ ¬supported(a)
           refuted                if
labelo(a)=refuted
           suspended              otherwise
```

The label is named **unrefuted**, not "accepted," deliberately: membership in the grounded extension means every attack on the node is currently defeated — survival under the criticism so far supplied, nothing stronger (P1, P2). The calculus has no stronger word to offer and refuses to imply one.

Orphaned $\neq$ false. Refuting a premise renders dependents `suspended_unsupported`, never `refuted`: losing one's ground is not being wrong. The distinction is load-bearing throughout §9.

Lemma 6.1 (Reinstatement). If k attacks a , j attacks k , and j is unattacked, then $\{j, a\} \subseteq G$. Reinstatement is derived, never ruled.

7. Machine invariants

N1 (No absorbing status). Every label admits an exit: `unrefuted` $\rightarrow$ `refuted` by new warranted attack; `refuted` $\rightarrow$ `unrefuted` by reinstatement; demonstrative refutations reopen via attack on their v ; rubric verdicts additionally via attack on their standard (Closure 2); support-lost artifacts recover when their premises do. No artifact — rules, standards, rulings, frame assertions included — is ever final (P1).

N2 (Perpetual proposability). ψ 's support is unbounded, and for every `unrefuted` a there exist proposable successors with strictly extended batteries (P10). §10 is N2's enforcement arm against *effective* collapse.

N3 (No insolubility). No rule, label, or scheduler state asserts that a problem cannot be solved. A problem leaves the frontier only by adjudicated retirement, translation into a successor, or resolution of its premises (§9.8); starvation of attention is a visible condition of the schedule, never a verdict on the problem. That problems are soluble is not a theorem the calculus proves; it is a state the calculus refuses to be able to deny.

8. Measures

Demarcation. $\text{crit}(a) \Leftrightarrow \text{interface.commitments} \neq \emptyset$; $\text{mod}(a) \Leftrightarrow$ the variation kernel $\mu(\cdot|a)$ yields inequivalent variants with positive probability; $\text{active}(a) \Leftrightarrow \text{crit} \wedge \text{mod}$. An artifact that forbids nothing has an empty attack surface and fails the root battery by program; an artifact whose mechanism is not load-bearing fails it equally (P4). Both die before testing, which is where most bad ideas should die.

Hardness to vary. Sample k bounded edits $a' \sim \mu(\cdot|a)$; $s(a)$ = the fraction that pass a 's battery while being battery-inequivalent to a ; $\text{HV}(a) = 1 - s(a)$. Where content parses structurally, μ **must** substitute at role level — swap the mechanism, the scope, the causal link — not merely reword (P9: a rename is the same explanation). The estimating battery excludes HV-type commitments (stratification: HV over a battery containing itself does not terminate). HV enters the graph only as a criterion inside problems or as a spawn signal — never as a gate on registration (C5).

Reach. Periodic budgeted cross-evaluation of unrefuted artifacts against *other* problems' criteria. A hit — the artifact accounts for material it was not built for, timestamped by the log to predate the encounter, on held-out material where available — is the strongest currency the calculus mints, and the promotion signal of §9.

Definition 8.1 (Knowledge, as a view).

$$\text{knowledge}(a) \Leftrightarrow \text{unrefuted}(a) \wedge \text{active}(a) \wedge \text{reach}(a) > 0$$

— information that has survived the criticism actually supplied, resists variation, and does work it was not built for. This is resilient information: the mark of knowledge as that which, once instantiated, tends to remain so and to propagate into problems it was not made for — here realized as survival-plus-hardness-plus-reach in the record. It is a view (C4) and an attention signal (C5),

never a status: promoting it to a label would reintroduce verification through the back door. And it is indexed: *unrefuted* means unrefuted by the attack supply so far, while the counterfactual ideal — resilience under all possible criticism — is approximated, never attained, through hardness-to-vary and the perpetual extension of batteries (N2).

9. Background: the standing layer

9.1 Two axes

Define **status** as above: truth-standing under criticism. Define **standing** as an artifact's role in the economy of generation: whether it is *framed* — one node among the packs' retrieved neighbours and precedents — or *does the framing*: rendered always, as the coordinate system every conjecture in a scope is written in or declared against.

A single-axis calculus cannot host P11.

Proposition 9.1 (Rigidity dilemma). Suppose frame role is a function of status, so that an accepted violation of a framework's commitments removes its frame role. Then either (i) framing toggles with the adjudication of each contested observation — and since observation-acceptance is revisable (P5, Closure 3), framing over the scope oscillates with every attack and reinstatement of the evidence, re-orienting every open problem in scope at each toggle; or (ii) the system prevents the toggle by delaying, suppressing, or immunizing — violating P3, P5, or P1 respectively. *Proof:* the alternatives are exhaustive. ■

The two-axis state is also what the logic of experimental tests demands. Refutation may be unilateral: one accepted violation ends a universal claim, and no rival need exist for it to do so. Theory-choice may not: a crucial test is intrinsically comparative,

requiring rival explanations of the total record — the apparatus's own account included — and a framework is rationally displaced only by a better explanation of everything it explained, its wounds now among the explananda. The status axis carries the first; the standing axis carries the second; a single axis must garble one into the other. The result is the ordinary condition of mature knowledge — **refuted and still framing** — which is precisely background held unproblematic *for the occasion* while remaining under open indictment.

9.2 Frame assertions; standing as a view

Definition 9.2 (Frame assertion). A frame assertion is an ordinary artifact *fa* whose content is a frame claim

```
{ subject b, scope  $\sigma$ , validity v, departure
  protocol }
 $\sigma$  : a total computable predicate over problem records
(C1: deterministic;
  embeddings may inform nomination, never
  membership)
v  $\in$  { universal, bounded(domain, tolerance) }
```

with interface refs: **mention** $\rightarrow$ **b**; **dependence** $\rightarrow$ **each reach record cited as its case**; mention $\rightarrow$ the wounds of any incumbent it succeeds. *fa* is **consulted** iff it is addressed to a promotion problem (§9.4) and $\text{final}(fa) = \text{unrefuted}$.

Definition 9.3 (Standing — derived, never stored).

```
standing(b)  $\exists$  background over  $\sigma \Leftrightarrow \exists$  consulted fa :
  subject(fa) = b  $\wedge$  scope(fa) =  $\sigma \wedge$  final(fa)
  = unrefuted
```

Instrument standing is not a third value: it is a consulted *fa* whose validity is **bounded** — the subject frames its granted domain as a

declared approximation, with the tolerance authored by its successor and attackable like anything (C3: the distinction is content, not type).

Law 9.4 (Mention law). A frame assertion MUST NOT carry a dependence ref on its subject. This single interface constraint is the whole separation of the axes: because fa merely *mentions* b, Pass 2 does not drag fa down when b is refuted — the wound does not touch the frame role. Because fa *depends* on its reach case, refuting that case cuts fa's support. Truth-standing and frame-standing are thereby decoupled at the level of edge roles, and both remain fully inside `att / dep`.

Whether a refuted subject *ought* to keep framing is itself a conjecture: attack fa directly. That argument is the succession or revocation case — adjudicated, never hardcoded.

9.3 Rent

Promotion is purchase of exposure. The subject's expectation set **is** its commitment set, and demarcation (§8) is the rent law: a candidate background must be active(b), with observation-valued commitments wherever its scope is empirical. Its commitments are held open indefinitely and checked against every relevant episode; the background tier is therefore the *most* falsifiable-in-practice position in the calculus, not a shelter (P11). A framework that cannot state what it forbids, or whose mechanism carries no weight, cannot frame — it fails the root battery before the question of standing arises. And promotion is an articulation event: a tacit framework becomes promotable only when factored into vocabulary, enumerated assumptions, and commitments — because the assumption ids are what departures declare against, and the commitments are what wounds violate. Articulation is not overhead; it is the manufacture of the attack surface.

9.4 Nomination and promotion

Nomination is a measure-rule over the log (channel (a) of C5): reach events for b spanning at least K_{frame} distinct problem lineages that jointly match a coherent candidate scope $\Rightarrow$ Spawn a **promotion problem**. The measure detects; it never decides.

Promotion is an ordinary Conj $\rightarrow$ Crit $\rightarrow$ Adj pass: a frame assertion is proposed addressing the promotion problem, whose criteria pin — beside the root battery —

1. *subject-demarcation* (program): active(b), observation-valued where empirical (§9.3);
2. *reach-integrity* (program over the log): the cited reach records exist; timestamps prove held-out standing where claimed;
3. *scope-determinism* (program): σ evaluates on problem metadata alone;
4. *compatibility*: an unrefuted consulted frame assertion overlapping σ routes this problem to **discrimination** against it (§9.7) — rivals never co-frame;
5. *accounts-for* (succession only): the candidate's subject covers the incumbent's standing fail verdicts, program-checked against the wound list with anchored-rubric backup.

Remark 9.5 (Default-consult closure). Grounded semantics labels the unattacked unrefuted, so an unchallenged frame assertion would otherwise frame its scope the moment it registered. Two native facts close the hole: the criteria above are instantiated at registration and generate demonstrative program warrants *before* the renderer's next consultation (Adj follows every registration); and the renderer consults only assertions addressed to promotion problems, which exist only by Spawn. A frame assertion born anywhere else is an ordinary artifact the renderer ignores.

Revocation requires no rule of its own: attack the reach records ("contaminated", "shallow", "not held-out") and fa loses support — $\text{final}(fa) = \text{suspended_unsupported}$ (§6), and the renderer stops

consulting it. Orphaned $\neq$ false does exactly the right work: revocation says *unearned*, not *wrong*.

9.5 Frame render semantics

The pack — the deterministic render each γ -call receives — comprises the problem and criteria, the target and its top attackers and defenders, the retrieved neighbourhood, the precedent slice, and, for every consulted fa whose σ matches the problem, the **frame slice**:

- the subject's articulation digest (compressed; expandable by view), and
- **the subject's standing attackers**. Wounds render in-frame, in every pack in scope.

The frame ships its own crisis. This is P11's searchlight with its hand shown: background conditions generation *and* displays its open indictments at the site of conditioning. Crisis is a render state, not a mode switch — the pressure toward rivals is applied at the only surface the calculus controls, namely what the generator is shown.

Departures. The frame slice carries a standing directive: *departures are permitted and must be declared*, as a list of the subject's assumption or commitment ids the candidate breaks with. Declaration removes the hidden-premise criticism's target (an undeclared conflict with the frame is criticizable as a silent assumption); the declaration is itself attackable (defend the assumption); and a departer that later reaches is routed by nomination to discrimination against the incumbent. Nothing scores departures, because no penalty channel exists to score them with — the freedom is by construction, not by rule. Scope predicates never read departure declarations: a departing conjecture cannot be exiled from the frame it is criticizing.

9.6 Wounds

A **wound** is nothing new: a fail verdict on one of the subject's observation-valued commitments $\Rightarrow$ demonstrative warrant $\Rightarrow$ attack edge $\Rightarrow$ Pass 1 computes refuted. The observation is itself on trial throughout (P5): its warrant's v carries an evidence ref, and an accepted attack anywhere in the evidence's dependence lineage collapses the warrant and reinstates the subject (Closure 3) — the faulty instrument, computed.

Proposition 9.6 (Wound persistence). A wound changes status(b) and does not change standing(b). *Proof:* the attack targets b ; fa carries no dependence on b (Law 9.4), so Pass 2 leaves fa 's label untouched; the renderer keys on $\text{final}(fa)$. ■

The failed verdict spawns a successor problem as always; under a consulted fa that successor is the **crisis problem** — a standing, addressable demand for an account of the wound — while the wound renders in-frame across the scope (§9.5). Newton between 1859 and 1915 is the intended model of this state: status-refuted, standing-background, perihelion on display in every pack, succession wanted.

9.7 Falls and succession

Standing ends in exactly two ways, and the two-pass labels distinguish them without new machinery:

Event	Mechanism	$\text{final}(fa)$	Cascade grade
Fall	succession lost, or a direct warranted attack on	refuted	premise-refuted

Event	Mechanism	final(fa)	Cascade grade
	fa sustained		
Revocation	reach case refuted $\Rightarrow$ support cut	suspended_unsupported	premise-unaccredited

Both end consultation; the labels differ, and §9.8 inherits the difference.

Succession is discrimination. Rival frame assertions over overlapping scope trigger the ordinary ≥ 2 -survivors discrimination spawn, resolved comparatively — as the comparative logic of §9.1 requires: pairwise ruling, cited decisive point, **mandatory order-swap** — the calculus's native symmetry instrument. One render exception is proper to succession: the succession pack **suppresses the incumbent's frame slice and renders both articulation digests**, so the trial of a frame is framed by neither party. The failure this mitigates deserves its name — *incumbent-judge bias*: a succession posed inside the incumbent's vocabulary is adjudicated by the defendant. The mitigation is symmetric exposure; a view from nowhere is not on offer.

Anomaly conservation. Nothing is deleted (P8), and the accounts-for criterion makes the successor claim the incumbent's wounds as its *own* commitments — attackably: the new frame must predict what broke the old one. Its scope statement must also fix the incumbent's residual validity domain; succession may therefore leave a residual bounded-validity frame assertion for the fallen subject, which thereby keeps framing its granted domain as a declared approximation — instrument standing (Def 9.3). The predecessor's domain of validity is authored by its successor, and that authorship is one more attackable claim.

9.8 Presupposition and the cascade

Problems posed under active background record the fact (Def 3.5: `provenance.frame`, deterministic at registration, registrant-editable at pose, immutable after). Presupposition is provenance: epistemically inert, steering spawn and attention, never labels — the frame can orient the searchlight without ever voting in court.

When a consulted fa leaves unrefuted standing (either grade), a replay program enumerates every problem carrying it and marks each **premise-orphaned**. Marks are lazily materialized: the orphan problem instantiates when its problem is next focused, and batch translation offers may materialize groups — the fall is one event; its thousandfold consequence is paid as the frontier is touched, not all at once. Pending marks deprioritize their problems in scheduling (attention only, C5).

Each orphan problem admits exactly three resolutions:

1. **retire** — the problem leaves the frontier, logged, never deleted: it died with its premise;
2. **translate** — a successor problem is posed in the succeeding frame's vocabulary, provenance recording the lineage:
succession lives on the problem layer;
3. **independence** — a finding that the problem never needed the premise; the orphan closes with that holding and the scheduler thereafter treats the problem as unorphaned (computed from the resolution; the problem's provenance is never mutated).
The independence-resolution rate doubles as the over-binding diagnostic on pose-time recording.

Fall-grade orphans carry *premise refuted*; revocation-grade carry *premise unaccredited*; neither auto-kills — orphaned $\neq$ false operates at the problem layer exactly as at the artifact layer, and none of the three resolutions is an insolubility verdict (N3): a

problem dies with its premise, moves to better language, or stands free of both.

Proposition 9.7 (Cascade totality). Every problem whose provenance carries fa receives an orphan mark when fa leaves unrefuted standing; no presupposing problem silently survives, and no mark resolves except by adjudicated work. *Proof:* marking is a total computable function of the log; resolutions exist only as registered problem-closures. ■

Here is the asymmetry P11 demands, now mechanical. An ordinary refuted conjecture retires nothing but itself, because nothing presupposes it. A fallen background's refutation *is* a premise-criticism of everything posed in its terms: one fall retires a thousand questions, translates a thousand more into a better vocabulary, and reveals some hundreds that never needed the premise at all – which is how the succession of frameworks becomes the succession of problems (P7).

9.9 Authority audit

Standing is render authority and nothing else. It is derived, never stored (C4); it is content and edge-structure, never a type (C3); it appears in packs and schedules, never in label computation (C5, §6); every object realizing it – the frame assertion, its reach case, its subject's commitments, the succession rulings – is attackable and reinstateable (N1, P6). Methodological privilege without epistemic privilege: the background frames every conjecture in its scope and can be dragged into court by any of them.

10. Generation and effective openness

The generator γ is a bounded pure function from packs to schema-valid candidates; it holds no state, adjudicates nothing, and controls no flow (P2: conjecture is unconstrained guessing; P6:

the guesser has no authority). Each call returns a *distribution* of candidates with stated typicality rather than a point — eliciting the tails that mode-seeking suppresses. Typicality is a statistic of the generator's distribution and nothing more: it prices attention and never weighs truth (C6). Every conjecture is born connected: packs carry the target's neighbourhood, and candidates are invited to declare relations, so unification is discovered by reach rather than designed (C5: coupling is a property of generation; reach is its measured shadow).

γ is a role, and the calculus is agnostic about its realization up to the contract. Nothing here presumes the generative side lies beyond computation: physical processes are computable, minds are physical, and a system that created explanatory knowledge would be a program. On the evolutionary account of knowledge there is no third ingredient — creation *is* conjecture and criticism — so this calculus is best read as the error-correcting half of exactly such a composite: selection formalized in full, generation constrained only in its *direction*. Problem-pressure, neighbourhoods, complements, and elicited tails say where variation is aimed; they are silent on how novelty arises. Absent a theory of that remaining half, the calculus licenses no claim in either direction: forced variation must not be advertised as creation, and creation must not be declared uncomputable. Both errors are refusals to say *we do not yet know*.

P10's honest gap: in-principle openness survives any conditioning, but a conditioned generator's *effective* support can collapse into a basin — and a frame slice is *deliberate, scope-wide conditioning*, the strongest the calculus ever applies. The countermeasures are attention-side by constitution: population structure in conjecture (persistent conditioning regimes diverging by their own lineages) with panmixia in criticism (one court; attacks cross lines freely); capture detection as replay programs over the log — stream contraction, attack-target entropy, criticism debt, reinstatement

rate, validity-node attack rate; responses that reweight rendering and budgets with hysteresis, never statuses. Promotion events are logged with before/after conditioning diagnostics: the capture cost of elevation is measured, not vibed. And because these instruments detect *stalled* dynamics, not *wrong-but-stable* ones, the calculus keeps a floor under its exogenous grounding ratio — the fraction of verdicts bottoming out in program checks, recorded evidence, and appellate rulings rather than closed-loop judgment. A background that is confidently, quietly wrong generates no wounds; only the anchors outside the loop bear on it.

11. Informal content

Informal domains enter through structure, never through types (C3). Candidates parse to a skeleton — claim, mechanism, scope, **forbidden cases** — and each forbidden case compiles at registration into a commitment: explanatory demarcation made real for prose (P4), with the judge's question shrunk from "is this good?" to "does case X violate clause Y?". The mechanism slot is load-bearing by the root battery, so a skeleton whose mechanism swaps freely is refuted by program, untested. Rubric evaluation resolves against **standard artifacts** — rubric text, evaluation mode, exemplars with holdings — which are ordinary artifacts: attackable, succeedable, and wired into Closure 2, so the productive attack in informal domains lands on the standard and its success reinstates every target judged under it, in one computed pass. Comparative rulings (anchored or pairwise, order-swapped, decisive-point-cited) replace absolute scoring; judge *behavior* is program-audited — paraphrase invariance, premise-deletion sensitivity, planted-flaw calibration — and audit hits enter as ordinary demonstrative warrants against the relevant validity nodes: formal machinery criticizing informal machinery with the full force of the graph. Human rulings enter as precedent artifacts

ranked first in render and attackable like anything: authority is pack ordering, never status privilege (P6). A share of the evidence corpus is registered sealed with scheduled reveals; a pass on revealed material is a reach hit with the strongest provenance the informal side can produce, the log's timestamps proving the artifact predates its evidence — the novel-fact criterion, mechanized.

12. Properties

Proposition 12.1 (Total computability). Every verdict is decidable within its declared budget (C2); Pass 1 is a least fixed point of a monotone operator on a finite lattice, polynomial in $|latt|$; Pass 2 is a single topological traversal of `dep`; standing is a query over labels; every view is a pure fold over the log (C1). No judgment in the calculus invokes an undecidable question. ■

Proposition 12.2 (No confirmation, no credence). The only evidence-bearing edges in the calculus ground attacks (Def 3.4, Closure 3): there is no derivation by which evidence raises any artifact's label, and no probability over any artifact's truth appears in label computation, measures, or scheduling weights on truth (C6) — generation statistics price attention only. What remains of "support" is exactly membership in the grounded extension with intact premises — survival (P2, P3), with the Popper–Miller decomposition standing guard against its probabilistic re-description. ■

Theorem 12.3 (No absorbing status). Under N1, every label of every artifact is revisable by a constructible registration: unrefuted falls to a fresh warranted attack; refuted reinstates by attacking the refuter, its validity node, its standard, or its evidence lineage; `suspended_unsupported` recovers with its premises. Frame assertions inherit all exits, and their subjects hold an additional

permanent exposure through open observation-valued commitments (§9.3). *Sketch:* each exit names a registrable object whose adjudication flips the label by §6; none is blocked by any guard, since guards are warrant-validity conditions only. ■

Proposition 12.4 (Axis independence). Status(b) can change without standing(b) changing (Prop 9.6), and standing(b) can change without status(b) changing (revocation attacks fa's reach case, not b). The axes are decoupled exactly by edge roles, inside `att / dep`. ■

Proposition 12.5 (Standing never adjudicates). Label computation reads `att` and `dep` only (§6); standing is consumed by render and schedule alone. Hence background confers no inferential weight: P6 and C5 are preserved under the standing layer, and P11's privilege is methodological in the strict sense. ■

Proposition 12.6 (Knowledge is a view). knowledge(a) (Def 8.1) is a total computable function of the log, consumed only by attention. It can rank, focus, and report; it cannot move a label. The characterization of knowledge as resilient, reaching information is thereby available to the machine without ever becoming a verification predicate. ■

13. Limits, stated

The calculus guarantees faithful bookkeeping — computable statuses, reinstatement, preserved error, replayable history, background that frames without ruling — and formalizes the selective half of the growth of knowledge while leaving the generative half as a role. That division is a statement of what has been theorized, not of what is possible: the whole process is computable in principle, and a generator that created explanations would complete the composite this calculus is built to correct.

Until such a theory exists, forcing variation is not producing creativity and is not advertised as such. Hardness-to-vary at small k is a spot-check standing on kernel-quality assumptions parked, visibly, in validity nodes. Informal verdicts are made narrow, comparative, precedent-anchored, and audited — not reliable. Capture instruments detect stalled dynamics; a consensus ossified around a shared blind spot is invisible from inside, and only the exogenous anchors bear on it, which is why the grounding floor is load-bearing rather than decorative. Nomination thresholds, scope predicates, slice budgets, and orphan scheduling are empirical constants; both failure directions are measurable from the log, and none is defended here. Succession neutrality is symmetric exposure, not neutrality. And a wounded background with no arriving rival frames forever — refuted, indicted in every pack, unreplaced, and never declared irreplaceable (N3). The calculus keeps the crisis visible and the succession problem open; it cannot force the successor into existence. That is not a gap in the machinery. It is what the growth of knowledge is like.